Hassan Ali Shah

Full Stack Developer | Software Engineer | Agentic AI Engineer

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PROFESSIONAL SUMMARY

Ambitious Full Stack Developer and Agentic AI Engineer with a strong foundation in Data Structures, Algorithms, and scalable system design. Experienced in building production-ready web applications and implementing machine learning models from scratch. Passionate about Generative AI and autonomous systems, with hands-on industry experience as an AI Engineer at startups. Internationally recognized innovator — represented Pakistan at the global stage in Act in Space 2026, organized by CNES (French Space Agency) and the European Space Agency (ESA). Proven leader with the ability to drive teams, solve complex problems, and rapidly adapt to emerging technologies.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, Dart, Swift, SQL

Frontend Development: React.js, JavaScript, Responsive UI Design

Backend Development: Node.js, Express.js, REST APIs, JWT Authentication, MERN Stack

Databases & Cloud: MongoDB, PostgreSQL, AWS

Mobile Development: Flutter, Dart, Swift

AI / ML: Generative AI, Agentic AI Systems, Logistic Regression, NumPy, Pandas, Data Preprocessing, Model Evaluation

Core CS: Data Structures & Algorithms, Object-Oriented Programming, System Design, Dynamic Memory Management

PROFESSIONAL EXPERIENCE

ML/AI Engineering Intern — CAASIAC 2024 – 2025

Contributed to end-to-end machine learning workflows at an AI-driven startup, gaining hands-on experience in a fast-paced production environment.

- Designed and optimized data preprocessing pipelines, improving model input quality and reducing training time.
- Developed and evaluated multiple ML models to support AI-driven problem-solving across business use cases.
- Collaborated with senior engineers on agentic AI research and implementation strategies for autonomous systems.
- Gained practical exposure to Generative AI tools and workflows in a real-world product environment.

AI/ML Intern — NeuroFive 2026 · 6 Weeks

Completed a 6-week internship focused on applied artificial intelligence and machine learning.

- Worked on AI/ML tasks spanning data preparation, model development, and evaluation.
- Applied machine learning concepts to practical, real-world problem statements in a team setting.
- Strengthened hands-on skills in Python-based ML tooling and AI workflows.

PROJECTS

Some of my Main Projects, rest can be found on GitHub

Full Stack Web Application (MERN Stack)

- Architected and deployed a production-ready web application featuring JWT-based authentication, dynamic real-time data flow, and a fully responsive UI using React.js, Node.js, Express, and MongoDB.

TaskFlow Pro – AI Task Management Platform (AI/ML + Full Stack) – Machine Learning from Scratch

- Built an AI-powered task management platform using Flask and Python with role-based dashboards, intelligent task assistance, automated feedback, and integrated code execution.

VORTEX OS – Custom Linux Shell Environment (Systems Programming)

- Engineered a modular Linux shell in Python with command execution, alias management, command chaining, persistent history, and a customizable theme engine.

Hassan-Verse – Interactive 3D Developer Portfolio (Frontend / 3D Web)

- Built an immersive 3D portfolio using React and Three.js, enabling interactive exploration of projects and skills through a virtual workspace. Deployed on Vercel with responsive, optimized rendering.

Cost-Based SQL Query Optimizer (Systems / Databases / C++)

- Implemented a cost-based SQL optimizer in C++ featuring recursive-descent parsing, Selinger join optimization, query rewrite rules, and an interactive SQL shell with EXPLAIN.

Uniflow – Classroom Management App (Flutter)

- Built Uniflow, a cross-platform mobile app inspired by Google Classroom, using Flutter and Dart to manage classes, assignments, and student-teacher communication.
- Designed core features including class creation, assignment posting and submission, and real-time notifications for a smooth academic workflow.

ACHIEVEMENTS & RECOGNITION

- **National Winner — Act in Space 2026:** Secured 1st place at the national level in Act in Space 2026, a prestigious space innovation challenge.
- **International Representative — Act in Space 2026 (France):** Selected to represent Pakistan at the international finals, organized by CNES (Centre National d'Études Spatiales) and the European Space Agency (ESA) — one of the few Pakistanis to compete at this global platform.
- **Team Lead — NASA Space Apps Challenge:** Led a team in the world's largest global hackathon organized by NASA, directing technical strategy and project delivery under a tight international competition timeline.
- **Participant — Code Rush Hackathon:** Competed in a high-intensity coding competition, applying problem-solving and algorithmic skills under pressure.

EDUCATION

BS Computer Science — Information Technology University (ITU), Lahore 2024 – 2028
CGPA: 3.02 / 4.0

FSc Pre-Engineering — Punjab Group of Colleges, Lahore 2022 – 2024
Percentage: 87%

Matric (Science) — Lahore Grammar School 2020 – 2022
Percentage: 92%

CORE STRENGTHS

- Strong analytical and problem-solving skills with a track record of competitive achievement.
- Ability to build robust systems from first principles — hardware-level to cloud-deployed applications.
- Proven leadership in high-pressure, international-level hackathon environments.
- Fast learner who thrives at the intersection of cutting-edge AI research and practical engineering.
- Deep grasp of core Computer Science fundamentals: algorithms, memory management, and system architecture.

Portfolio Link

You can find my LinkedIn and my GitHub profile through my portfolio

https://hassan-ali-17.github.io